Republic of the Philippines
Sorsogon State University
COLLEGE OF INFORMATION
AND COMMUNICATIONS TECHNOLOGY
Bulan Campus
Zone 8, Bulan, Sorsogon
Tel. No.; 056 311-0103; Email Address: cict@sorsu.edu.ph

BAGONG PILIPINAS

CC 101 Laboratory Activity: Exploring the Windows Command Line

Item	Details
Course	CC 101 Introduction to Computing (BSCS 1st Year)
Course Outcome	CO2: Identify the functions and relationships of major hardware and software components
Output	Command Line Exploration Documentation (PDF)
Estimated time	2 to 2½ hours
Reference	CC 101 Lecture Notes: Exploring the Windows Command Line

Instructions

Follow every step in order, take the screenshots marked **[Screenshot]**, and answer the questions marked **[Answer]** in your documentation.

Before you start

- You need a Windows 10 or 11 computer. No administrator rights are needed.
- To take a screenshot, press **Windows + Shift + S**, drag over the terminal window, then paste it (Ctrl + V) into your documentation.
- Replace the sample values (Juan Dela Cruz, BSCS 1-1) with **your own** details.
- Type every command yourself. Do not copy and paste; typing is how you learn.

Part A: Open the Two Shells

1. Press Windows + R, type cmd, and press Enter. Type cls and press Enter.
2. Type echo Juan Dela Cruz, BSCS 1-1 (your own name and section) and press Enter. **[Screenshot] Figure A1**
3. Press Windows + R, type powershell, and press Enter. Type the same echo command, but put the text in quotation marks: echo "Juan Dela Cruz, BSCS 1-1". PowerShell treats the comma as a list separator without them. **[Screenshot] Figure A2**

[Answer] A-Q1. Compare the two prompts. What is different, and what does each part of the Command Prompt prompt mean?

Part B: Inspect Your System (Command Prompt)

Use the Command Prompt window for Parts B to D.

1. Run these commands one at a time: whoami, then hostname, then ver. **[Screenshot] Figure B1**
2. Run the command below. It may take 10 to 30 seconds. **[Screenshot] Figure B2**

```
systeminfo | findstr /B /C:"OS Name" /C:"OS Version" /C:"System Type" /C:"Total Physical Memory"
```

3. Run ipconfig. Find the adapter that shows an IPv4 Address. **[Screenshot] Figure B3**

[Answer] B-Q1. Complete this table in your documentation using your results:

Item	Your result

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Item	Your result
Logged-in account (whoami)	yhamei\acer
Computer name (hostname)	yhamei
OS name	Microsoft Windows 11 Home Single Language
OS version	10.0.26200 N/A Build 26200
System type (x64 or ARM)	x64-based PC
Total physical memory (RAM)	7,541 MB
IPv4 address	10.15.241.90

[Answer] B-Q2. The | symbol in step 2 is called a **pipe**. Run systeminfo alone and compare. In your own words, what did | findstr do?

Part C: Build a Folder Structure

1. Go to your user folder and create the lab folders:

```
cd %USERPROFILE%
mkdir CC101-Lab
cd CC101-Lab
mkdir Notes Projects Screenshots
```

2. Create a text file about yourself (use your own details):

```
echo Name: Juan Dela Cruz > profile.txt
echo Section: BSCS 1-1 >> profile.txt
echo Favorite app: YouTube >> profile.txt
type profile.txt
```

[Screenshot] Figure C1 (show all commands and the output of type)

3. Run tree /F to display your folder structure. [Screenshot] Figure C2
4. Open File Explorer, go to C:\Users\(your account)\CC101-Lab, and open profile.txt in Notepad. [Screenshot] Figure C3 (File Explorer and Notepad side by side)

[Answer] C-Q1. What would happen if you used > instead of >> in the second echo line? Test it on a new file (test.txt) and explain.

Part D: Copy, Rename, and Delete Files

```
copy profile.txt Notes\profile-copy.txt
cd Notes
ren profile-copy.txt about-me.txt
echo This file will be deleted > temp.txt
dir
del temp.txt
dir
cd ..
tree /F
```

[Screenshot] Figure D1 (from copy to the first dir) and [Screenshot] Figure D2 (from del to tree /F)

[Answer] D-Q1. After del temp.txt, check your Recycle Bin. Is temp.txt there? What does this tell you about deleting files from the command line?

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Part E: Explore PowerShell

Switch to the PowerShell window.

1. Go to your lab folder and list and read its files:

```
cd $HOME\CC101-Lab
Get-ChildItem
Get-Content profile.txt
$PSVersionTable
```

[Screenshot] Figure E1

2. Run Get-Help Get-ChildItem. If PowerShell asks whether to run Update-Help, type N and press Enter. [Screenshot] Figure E2
3. Take the **last three digits of your student ID number** (for example, 2025-00347 → 347). Convert it with PowerShell:

```
[Convert]::ToString(347, 2)
[Convert]::ToString(347, 8)
'{0:X}' -f 347
```

[Screenshot] Figure E3

[Answer] E-Q1. Write your number in decimal, binary, octal, and hexadecimal. Keep this; you will verify it by hand in Week 5.

Part F: Discover a New Command

1. In Command Prompt, type help to see a list of commands.
2. Choose **one** command that was **not** taught in this lesson and is safe to run (for example: date /t, time /t, tasklist, vol, or color 0A).
3. Read its help with command /? (for example, tasklist /?), then run it. [Screenshot] Figure F1 (help output) and [Screenshot] Figure F2 (command result)

[Answer] F-Q1. What does your chosen command do, and when would it be useful?

Command chosen: tasklist

What it does: It shows all programs and processes currently running on your computer. It lists each program's name, process ID, session details, and how much memory it uses. When useful: It helps you check what is running, find programs that are frozen or using too much memory, and confirm if a program has fully closed after you exit it.

Part G: Reflection

[Answer] Answer in 3 to 5 sentences each:

1. G-Q1. Which task in this activity was faster in the CLI than in the GUI? Which was easier in the GUI? Explain why.

   Creating folders and files was faster in the CLI because I could do it instantly with short commands instead of clicking through many windows. The GUI was easier when viewing and organizing files, since I could see everything visually and drag items around without needing to remember exact commands. CLI is quicker for tasks you know well, but GUI feels more

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natural and forgiving when you are still learning.

2. G-Q2. What error did you encounter (if any), what caused it, and how did you fix it?

I encountered errors when I tried Command Prompt commands inside PowerShell — they work differently and did not run. I also accidentally created nested folders and typed labels directly into the command line, which caused more errors. I fixed these by switching to the correct Command Prompt window, double-checking my current folder path, and only typing the actual commands exactly as shown.

3. G-Q3. Give one real-world job or situation in computing where knowing the command line would be important.

Knowing the command line is very important for IT professionals and system administrators who manage many computers at the same time. Instead of doing tasks manually on each machine, they can use commands to automate work, fix problems faster, and control systems remotely. This saves hours of time and ensures the same task is done consistently across every computer.

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Documentation and Submission

Submit one PDF that records what you did, what the computer showed, and what you learned. Someone reading it should be able to repeat your work.

Required Contents (in this order)

1. **Cover page:** activity title, your full name, student ID, section, date performed, and instructor's name.
2. **Introduction** (3 to 5 sentences): what the command line is and the purpose of this activity, in your own words.
3. **Procedure and results, Parts A to F:** for each part, a short heading, the commands you typed, the labeled screenshots, and your answers to that part's [Answer] questions.
4. **Commands learned table:** every command you used and what it does, in your own words.
5. **Reflection:** your answers to G-Q1 to G-Q3.

Formatting Rules

Item	Requirement
File type	PDF (write in Word or Google Docs, then export as PDF)
File name	CC101_CLI-Lab_Lastname_Firstname.pdf
Font	Arial or Calibri, 11 pt; commands in Consolas or Courier New
Screenshots	Crop to the terminal window; text must be readable. Label each one below the image, e.g., "Figure B2. Filtered systeminfo output"
Personal details	Your own name, section, and student ID must appear in Figures A1, C1, and E3
Total screenshots	16 (A1–A2, B1–B3, C1–C3, D1–D2, E1–E3, F1–F2)

Sample Commands Learned Table

Command	Shell	What it does
mkdir Notes	Command Prompt	Creates a folder named Notes in the current folder
type profile.txt	Command Prompt	Displays the contents of profile.txt

Submission

- Upload the PDF to the activity page in the course LMS before the deadline your instructor sets.
- Outputs must be your own work. Identical screenshots or answers will be treated as plagiarism under the class policy.
- Keep your CC101-Lab folder; later activities may build on it.

Grading Rubric

The documentation is scored out of 100 points and counts under Laboratory: Project.

Criterion	Points	Full marks	Partial marks	Low marks

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Criterion	Points	Full marks	Partial marks	Low marks
Completeness of procedure	30	All 16 labeled, readable screenshots; every part A to F documented	1 to 4 screenshots missing or unreadable	5 or more screenshots missing
Accuracy of results	20	System-information table complete; commands and conversions correct and personalized	Minor errors or 1 to 2 missing values	Many errors, or results not the student's own
Answers to guide questions (A to F)	20	Correct, clear, in the student's own words	Mostly correct but brief or vague	Missing, incorrect, or copied
Reflection (G-Q1 to G-Q3)	15	Specific, honest, and linked to the activity	General statements with few specifics	Missing or off-topic
Organization and formatting	15	Follows the required order, naming, and formatting rules; commands learned table complete	Some rules not followed	Disorganized; wrong file type or name
Total	100			

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